Diagnose Before Solving: AI Simulation

Interface Design (v1.1)

Project owner: David Bacon
Document status: Revised for publication
Date: August 31, 2026 (v1.0); revised November 12, 2026 (v1.1)
Deliverable: 6a of 11, inserted after the Technical Design
Depends on: Scope Document, Design and Prompt Architecture, Evaluation Design, Technical Design

> **Revision note (v1.1, November 12, 2026).** Two references to a turn counter are removed from sections 8 and 10; the counter itself was removed in D-081 and section 3.2 already said so. A cross-reference in section 3.2 is corrected. Sections 8 and 9 now describe a generic host frame rather than a Storyline slide, since no Storyline module embeds the simulation (D-173, D-188). No interface decision changed.

---

1. Purpose

The preceding documents specify what the simulation does in considerable detail and never say what it looks like. This one closes that gap: the framing, the layout, the states, and the visual system.

It matters more than a spec of this size usually would, for a specific reason. The primary audience is a hiring manager who will spend two to five minutes with the artifact (Scope 2.1). Whatever they conclude about the design work will be formed largely by what they see, before they read a word of the supporting documents.

---

2. The framing decision

2.1 The tension

The fiction is a phone call. Marcus called the support line, he speaks rather than writes, his responses are specified as speech, and his opening line is something a person says aloud. The interaction is typed text.

A messaging-app interface is the obvious answer for familiarity, and familiarity is worth a great deal here: a visitor with two minutes should not spend any of them working out how to use the thing. Chat bubbles are understood by everyone without instruction.

But taken literally, a messaging UI contradicts the scenario. If it looks like SMS, then Marcus called and is now texting, and his three-to-five-sentence replies read as oddly long messages rather than as normal speech.

2.2 Decision: a live call, shown as a running transcript, using messaging conventions

The interface keeps every familiar mechanic of a messaging app (aligned bubbles, a text input at the bottom, a typing indicator, vertical scroll) and frames the container as a call in progress rather than a message thread.

Concretely, that means a call header above the conversation, no timestamps, no read receipts, and no messaging chrome that implies asynchronous exchange.

This is not a compromise between two options; it uses the familiar mechanic while keeping the fiction that makes the content work. Three things follow from it that a pure messaging treatment would lose:

- **Marcus's long replies become natural.** A five-sentence turn is unremarkable as speech and slightly strange as a text message. Framing the bubbles as transcript rather than SMS resolves the whole length question.

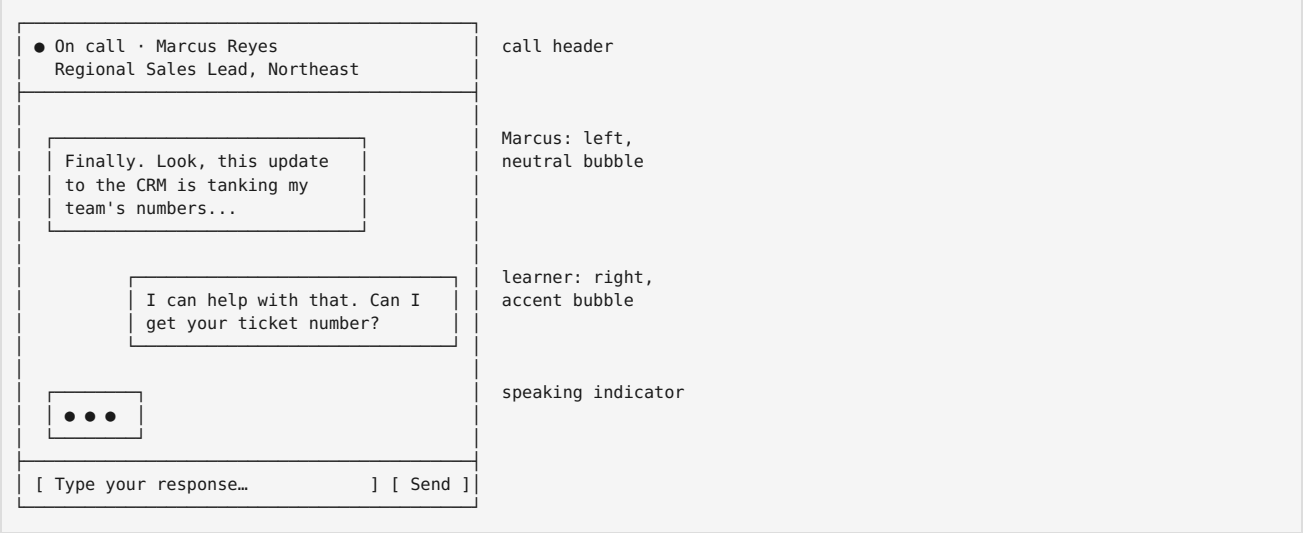
- **The header does instructional work.** "On call with Marcus Reyes, Regional Sales Lead, Northeast" establishes who he is and why his opinion carries weight, in a place the learner will look before they read anything else.
- **Urgency is implied by the form.** A call is synchronous and someone is waiting. A message thread is not, and a learner who feels they can take their time is in a different exercise than the one designed.

2.3 The option not taken

A support-console treatment (ticket panel, customer record, canned responses) would be more authentic to what an internal support specialist actually looks at all day. It was rejected because it trades the artifact's most valuable property, immediate legibility to a non-specialist visitor, for realism that only a support professional would recognise. It would also add furniture the simulation does not use: this scenario has no ticket to display and no record to consult, so the panels would be decoration.

---

3. Conversation view



3.1 Components

Element	Specification
Call header	A live indicator dot, "On call," Marcus's name, and his role on a second line. Fixed at the top; does not scroll away. Collapses to a single line below 480px or in a short frame
Marcus's turns	Left-aligned, neutral surface, rounded. First turn preceded by his initials in a circle. No photograph (section 3.4)
Learner's turns	Right-aligned, accent surface. No avatar
Speaking indicator	Three animated dots in an empty left-aligned bubble while a response is generating. **No "typing" label**, which would contradict the call framing
Input	Single-line field that grows to a maximum of four lines. Enter sends; Shift+Enter adds a line. A visible Send button, because Enter-to-send is invisible to a first-time visitor and the button is the only affordance on touch
Scroll	Auto-scrolls to the newest turn. Manual scrolling upward is never interrupted by an arriving message

3.2 Deliberately absent

- **Timestamps.** They imply an asynchronous thread and add visual noise to every turn.
- **Read receipts and delivery states.** Same reason, and they are meaningless in a call.
- **Emoji reactions, attachments, voice-note affordances.** Furniture the simulation does not use.
- **Any rubric, criteria list, or coaching hint.** Decided in D-055 and load-bearing: a visible criterion turns handling a person into completing a checklist.

- **An anger meter or emotional-state display.** Decided in D-025. It would turn the conversation into a game to be optimized.
- **A turn counter.** Decided in D-081. Counters do not exist on real calls, and part of the skill being practised is reading how much patience you have left without a number telling you. The 15-turn cap is still enforced server-side; the learner simply cannot see it. The signal that a call is running long is delivered by Marcus instead (section 3.3).

3.3 The duration signal replaces the counter

Removing the counter (D-081) removes the learner's only warning that the call is finite, and without a replacement the turn-limit ending would arrive from nowhere. The replacement is diegetic: **Marcus himself signals that the call is dragging.**

From roughly turn 11, his dialogue can register the length of the call: asking where this is going, mentioning he has a pipeline review, observing that they are going in circles. A real caller does exactly this, which makes it better than a counter rather than merely a substitute for one.

It changes what he says, not what he feels. The signal is surface tone, not a move up the ladder. It does not change his `intensity`, it is not caused by anything the learner did wrong, and it is never counted as an escalation event. Those constraints keep D-035 (only the learner's behavior moves Marcus) and D-025 (an escalation requires an identifiable escalating move) intact. The full specification is in the Design and Prompt Architecture document, section 6.3.

3.4 On not using a photograph

Marcus is represented by initials in a circle, not a face. A stock photograph would attach a real person's likeness to a fictional character who behaves badly, and an AI-generated face invites the viewer to evaluate the image instead of the writing. Initials cost nothing and carry no such risk.

---

4. States

4.1 The five states of the main view

State	What the learner sees
Landing (standalone only)	Scenario, the learner's role, what a good outcome looks like in general terms, the note that this is normally preceded by course instruction, and a Start button. Suppressed when `?embed=1` (Scope 4.1, D-055)
In call	Section 3
Evaluating	Input disabled. A system line confirms what happened ("Marcus has ended the call"), then a visible progress indicator. This is the longest wait in the application (D-063), so it must never look frozen
Results	Section 5. Replaces the conversation entirely
Transcript review	Section 6. Read-only conversation with cited turns highlighted

4.2 System lines

Between-turn events that are not dialogue (the call ending, a warning that the session expired) appear as centered, quiet, non-bubbled text. They read as stage direction, clearly outside the conversation, and are never styled as something Marcus said.

---

5. Results view

Replaces the conversation and takes the full frame (D-026). It should look like a report, not a chat, because it is structured content rather than conversation.

```
┌──────────────────────────────────────────┐
│  Call ended · Marcus escalated           │
│                                          │
│         38 / 100        Not yet passing  │
│                          70 needed       │
├──────────────────────────────────────────┤
│  How you scored                          │
│                                          │
│  Acknowledged before diagnosing   13 / 25  │
│  ▸ Partial                               │
│  Asked open diagnostic questions  25 / 25  │
│  ▸ Full                                  │
│  ...                                     │
│                                          │
│  Deductions                              │
│  Promised beyond your authority      −15   │
├──────────────────────────────────────────┤
│  What happened                           │
│  What worked                             │
│  What to do differently                  │
│  Where Marcus escalated                  │
│  The one thing                           │
├──────────────────────────────────────────┤
│  [ Review the call ]      [ Try again ]  │
└──────────────────────────────────────────┘
```

Ordering is deliberate. The outcome and score come first because that is what a learner looks for and withholding it does not create suspense, only irritation. The criteria breakdown comes next as the bridge between the number and the prose. The narrative feedback carries the weight and gets the most room (E4: the feedback is the deliverable, the score is the summary).

Criterion rows show points, level, a one-line reason, and the evidence quotation. Expanded by default on a large frame, collapsed below the breakpoint where they would push the narrative feedback off screen in a 405px-tall frame.

Deductions appear as their own group, not mixed into the criteria, because subtractions and awards are different kinds of thing and mixing them makes the arithmetic hard to follow.

Escalation moments cite their turns and link into transcript review.

Both actions are always available. Review the call, and Try again (D-070).

---

6. Transcript review

Reached from the results view. The conversation, read-only, with every turn cited in the feedback highlighted and labelled with what it was cited for ("criterion 2, full credit" / "escalation, turn 2").

This exists because the feedback quotes specific turns, and telling a learner what they said at turn 7 without a way to see turn 7 is a dead end. It is nearly free to build: the evaluator already returns turn indices for every judgment and every escalation event (D-025, D-026).

A persistent control returns to the results view. The learner can move between the two freely.

---

7. Demo mode and error states

Demo mode (D-024) is not an error page and should not look like one. It explains the situation in plain language, encourages trying again later, and offers the recorded playthrough as a clearly primary action.

Once opened, the demo shows the recorded conversation in transcript-review styling, with a banner marking it as a recorded session rather than a live one. The banner is not dismissible; a visitor should never be confused about whether they are talking to Marcus.

The scoring-failed state (D-064) says plainly that scoring did not complete and offers transcript review and a retry. It shows no number. A score presented uncertainly is worse than no score.

---

8. Visual system

Restrained and professional. This is an enterprise learning artifact, not a consumer app, and it may sit inside a host page, such as a course slide, whose design it should not fight.

Aspect	Specification
Type	One family, system stack for reliability inside an iframe. Conversation at 15 to 16px; nothing below 13px
Color	Neutral greys for surfaces and Marcus's bubbles; a single accent for the learner's bubbles and primary actions; one positive and one cautionary tone reserved for scoring. No gradients
Scoring colors	Used sparingly and never alone. Pass and fail are always stated in words as well (Scope 7)
Density	Generous line height in the conversation; tighter in the results view, which carries more structured content
Motion	Only the speaking indicator and a brief fade as new turns arrive. No slide-in animations; they cost attention and add nothing
Theme	**Single light theme.** No dark mode, and no automatic switching on system preference

On the single theme. Inside a host page the simulation is one element of a composed design, and a panel that unexpectedly turns dark because of a viewer's operating system setting would clash with a page built against a light layout. A theme that changes underneath the page author is a defect from their point of view. One well-made light theme, designed to sit neutrally on a page, is the right call, and it removes an entire category of things that can look wrong.

---

9. Frame and viewport behavior

The application must work in three quite different rectangles: a desktop browser, a phone, and a frame on a host page (Technical Design section 10).

Context	Behavior
Desktop standalone	Conversation column capped around 720px and centered. Full-width text is hard to read and looks unconsidered
Phone, 375px	Single column, full width. Input pinned above the keyboard. Header collapses to one line
16:9 host frame at 1280x720	Comfortable. The recommended target for any 16:9 host such as a Storyline slide (D-023, Scope 6.3)
16:9 host frame at 720x405	Workable but tight. Header collapses, criterion rows collapse, results view scrolls within its own container

The page never scrolls horizontally, in any context. The conversation and the results view each scroll vertically inside their own container rather than growing the page, which is what keeps the layout stable inside a fixed-size frame.

---

10. Accessibility

Implements the Scope section 7 baseline; verified by test Suite G.

- Speaker is conveyed by alignment **and** a name label, never by bubble color alone.
- The conversation is an ARIA live region so new turns are announced as they arrive.
- Focus moves deliberately when the view changes, and never lands on a control that no longer exists (D-026).

- **Focus returns to the input the moment Marcus finishes replying.** The input is disabled while a reply is in flight, and a disabled element loses focus; re-enabling it does not restore it. Without an explicit return, the learner must click back into the field after every turn. Focus returns only while the call is live; at a terminal state it belongs on the results view.
- The whole session is completable by keyboard, including sending turns and reaching results.
- All text meets WCAG AA contrast, including quiet secondary text such as the header's role line and the system lines, which are the elements most likely to fail.
- The speaking indicator has a text equivalent for screen readers.

---

11. Open questions

All three resolved.

1. ~~**Does Marcus's role line belong in the header?**~~ **Yes, kept in the header, and it is the first thing to collapse when the frame is short.**
2. ~~**Should the turn counter change appearance as it runs low?**~~ **There is no turn counter.** See D-081 and section 3.3.
3. ~~**Criterion rows expanded or collapsed by default?**~~ **Expanded on a large frame**, collapsed below the breakpoint where they would push the narrative feedback off screen. On a desktop the expanded state shows the depth of the assessment, which serves the portfolio audience directly.